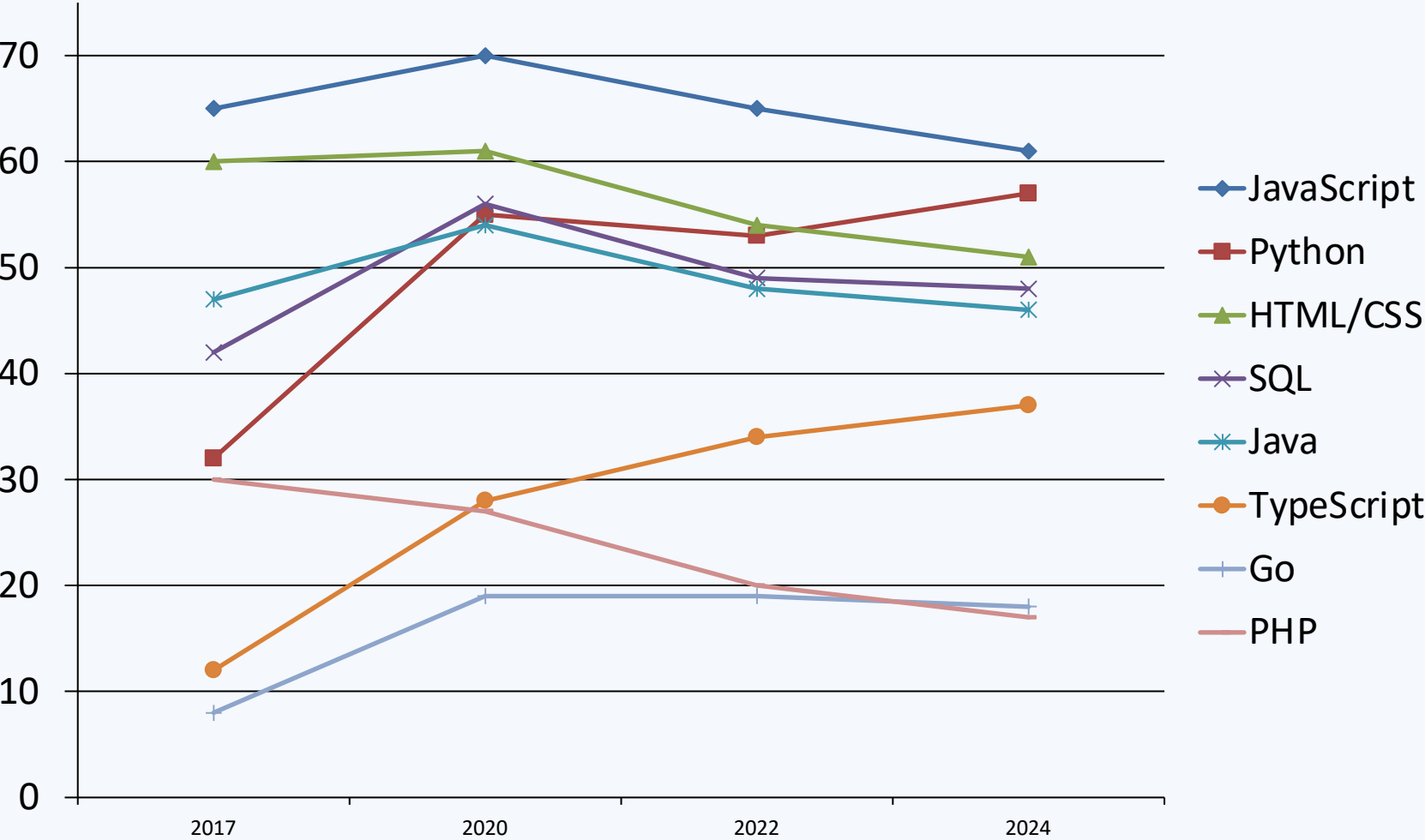


State of Developer Ecosystem 2024: Language Trends,

A data-driven report for engineering leaders, CTOs, and developer experience teams

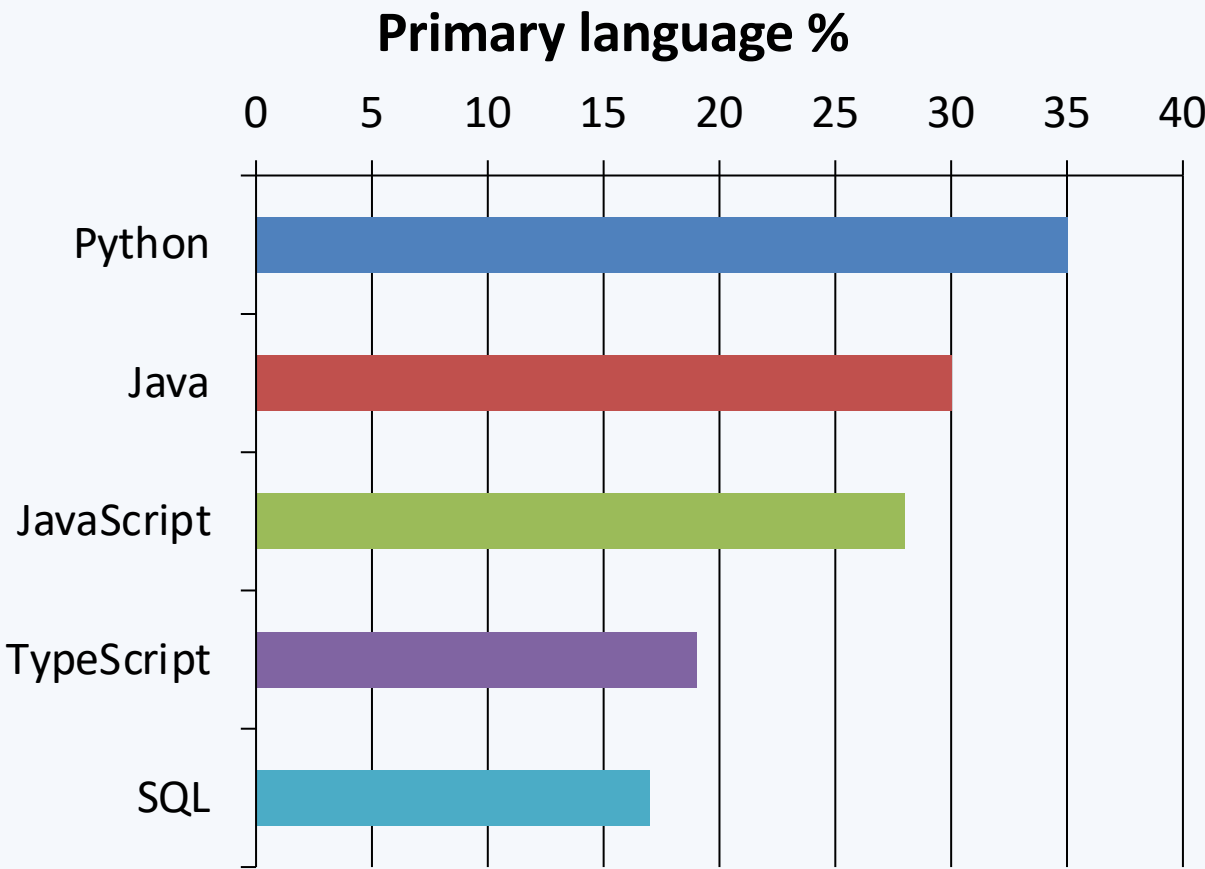
Based on JetBrains' annual survey of 23,262 developers worldwide, covering programming languages, tools, AI attitudes, and career trends.

Python Closes the Gap: 8 Years of Language Usage Shifts



- JavaScript peaks at 70% (2020), then declines to 61% (2024).
- Python rises from 32% to 57%, nearly matching JavaScript.
- TypeScript triples from 12% to 37%.
- PHP falls from 30% to 17%.
- Top-5 landscape is reshuffling.

Python Is Now the #1 Primary Language at 35%



Primary language share

- Python 35%
- Java 30%
- JavaScript 28%
- TypeScript 19%
- SQL 17%

Adoption intent (new primary) remains low

- Python 6%
- TypeScript 5%
- Java 4%
- SQL 4%
- Ecosystem lock-in: most developers stay in-stack

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Interpretation

- Type-safe, modern-tooling languages dominate top ranks.
- TypeScript (199) and Rust (198) are nearly tied.
- Python remains strong at 151, balancing versatility and usability.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
Data Science / Analytics	Python	28%
User Interface	TypeScript	62%
User Interface	JavaScript	58%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%

What This Means for Your Technology Strategy in 2025

- **Invest in Python for AI/ML teams**

Python leads AI/ML (31%), data science (28%), and data processing use cases.

- **Make TypeScript a core frontend/full-stack standard**

TypeScript leads UI (62%) and is the highest-satisfaction language (199).

- **Use Rust and Go for infrastructure bets**

Rust satisfaction is #2 (198); Go remains a key modern systems option (18% usage).

- **Plan for legacy-stack hiring constraints**

PHP declines from 30% to 17% (2017-2024); Ruby declines from 10% to 4%.

- **Prioritize developer experience in stack choices**

Higher satisfaction aligns with type safety and modern tooling.